RWANDA TVET BOARD

Ministry of Education | Republic of Rwanda

PROJECT PROPOSAL

RTB Schools GIS Monitoring & Intelligence System

Drone Imagery • KMZ/KML Geospatial Processing • 3D Visualization • Population Analytics

Prepared For	Rwanda TVET Board (RTB), Ministry of Education
Project Type	Geospatial Intelligence & Infrastructure Monitoring Platform
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1. Executive Summary

The Rwanda TVET Board (RTB), established under Presidential Order No. 123/01 of 15/10/2020, is mandated to regulate and promote Technical and Vocational Education and Training (TVET) from Level 1 to Level 5 of the Rwanda Education Qualification Framework. RTB currently oversees hundreds of TVET schools distributed across all five provinces, 30 districts, and 416 sectors of Rwanda — with the national goal of placing at least one school in every sector.

Despite this expansive footprint, RTB's current infrastructure monitoring capabilities are limited by reliance on conventional Google Maps imagery, which suffers from low resolution, imagery lag, and an inability to capture granular ground-level structural conditions. Critical planning decisions — whether to renovate, upgrade, expand, or build anew — are being made without adequate geospatial intelligence.

This proposal presents the RTB Schools GIS Monitoring & Intelligence System: a purpose-built, nationally scalable geospatial decision-support platform that integrates:

- High-resolution drone-captured orthomosaic imagery for each school site
- KMZ/KML geospatial file processing and 3D rendering via CesiumJS
- PostGIS-backed spatial analysis with buffer zones, terrain classification, and proximity analytics
- Live population density integration via the ArcGIS FeatureServer population API
- A rule-based Decision Engine that automatically scores and classifies each school by priority level and recommended intervention

The platform transitions RTB from reactive, manual school monitoring to proactive, data-driven infrastructure governance — directly supporting Rwanda's Vision 2050 and the National Strategy for Transformation (NST-1) pillars of quality education and digital transformation.

2. Problem Statement & Background

2.1 Institutional Context

RTB's Training Management Department is charged with standards compliance, school infrastructure development, and the operational oversight of TVET providers nationwide. According to RTB's Strategic Plan 2021–2024, data on TVET schools must be disaggregated by provider, district, sector, and cell — yet no unified digital platform currently enables this with geospatial precision.

A student survey indicated that only 58% of TVET schools met established quality standards. With over 282 of Rwanda's 416 sectors having at least one TVET school, the monitoring burden is substantial and growing.

2.2 Core Problems Identified

#	Problem	Impact
1	Low-resolution Google Maps imagery	Inability to assess structural conditions remotely
2	No centralized school geospatial registry	Decisions made on incomplete location data
3	No population-capacity comparison tool	Misallocation of infrastructure investment
4	No terrain/risk classification system	Schools in hazardous zones go undetected
5	No structured renovation priority system	Delayed response to urgent infrastructure needs

6	KMZ data from Google Earth not systematically utilized	Rich geospatial data wasted or siloed

2.3 Strategic Alignment

This system directly serves RTB's mandate under Strategic Outcome 7 (TVET Provision Mapping) and aligns with Rwanda's NST-1 Priority 4 — Quality Education — and the ICT Sector Strategic Plan 2018–2024, which emphasizes government digital transformation and data-driven governance.

3. System Overview & Key Features

3.1 System Architecture Summary

The platform consists of four integrated layers:

Layer	Components	Purpose
Presentation Layer	React + TypeScript + TailwindCSS, Leaflet, CesiumJS	2D dashboards and 3D terrain visualization
API Layer	Node.js (NestJS), RESTful APIs	Business logic, KMZ processing, routing
GIS Engine	PostGIS, GDAL, ogr2ogr, JSZip, xml2js	Spatial queries, KMZ parsing, geometry storage
Data Layer	PostgreSQL + PostGIS, ArcGIS FeatureServer API	Schools DB, population data, drone tiles

3.2 National Schools Dashboard (Module 1)

The home screen provides RTB leadership with a national overview of all TVET schools across Rwanda's five provinces. Key features include:

- Interactive national map (Leaflet) with schools plotted by GPS coordinates, color-coded by priority classification
- Province/District/Sector filter panel for hierarchical navigation
- KPI tiles: Total schools, schools below standard, schools flagged for renovation, high-risk locations
- Bar and donut charts showing distribution by school type, condition, and urgency level
- Search and sort functionality by school name, district, condition score, or last inspection date

- Export reports to PDF or Excel for stakeholder distribution

3.3 Individual School Monitoring Dashboard (Module 2)

On selecting any school, the user enters a detailed monitoring view with the following sub-modules:

3.3.1 School Map Viewer

- Displays the school's Google Earth KMZ file rendered as a high-resolution interactive map
- Toggle between: 2D satellite view (Leaflet), 3D terrain mode (CesiumJS), and drone orthomosaic overlay
- Polygons from KMZ file rendered with building footprints, boundary demarcation, and access routes
- Surrounding context: roads, rivers, hills, and adjacent settlements from OpenStreetMap and terrain data

3.3.2 KMZ File Upload & Management

- Drag-and-drop interface to upload .kmz files directly from Google Earth
- System automatically unzips KMZ, parses embedded KML (XML), extracts coordinates, converts to GeoJSON
- Parsed geometry stored in PostGIS database linked to the school record
- Version history: track changes when new KMZ files are uploaded for the same school
- Validation alerts if KMZ file is malformed or contains incomplete geometry

3.3.3 School Details Registration Form

A structured data entry form capturing all parameters required for decision-making:

Category	Data Fields
Identification	School name, type (VTC/TSS), ownership (public/private), year established, RTB license number
Location	Province, District, Sector, Cell, Village, GPS coordinates (lat/long)
Infrastructure	Number of classrooms, workshops, dormitories, latrines, admin blocks; condition of each (Good/Fair/Poor)
Capacity	Student enrollment (male/female), max capacity, trainer count, current utilization rate
Site Conditions	Land area (hectares), topography (flat/hilly/slope %), flood risk, landslide risk, road accessibility
Utilities	Water access (Yes/No/Source), electricity (grid/solar/none), internet connectivity
Age & Condition	Year of last renovation, structural assessment score (1–10), roof condition, drainage status

4. KMZ Geospatial Processing Pipeline

KMZ (Keyhole Markup Language Zipped) is the standard export format from Google Earth and represents a zipped KML file containing XML-based geospatial data — including coordinate geometries (points, lines, polygons), metadata, styles, and optional image overlays. The system implements a production-grade 7-step pipeline:

Step	Operation	Technical Implementation
1	Upload KMZ File	Frontend file input (React) with MIME type validation (.kmz). File streamed via multipart form-data POST to NestJS API endpoint.
2	Unzip KMZ	JSZip library extracts embedded KML file from the ZIP archive. Handles multi-file KMZ packages with overlay images.
3	Parse KML (XML)	xml2js parses KML XML into JSON object tree. Extracts all Placemark nodes containing Point, LineString, and Polygon geometries.
4	Extract Geometries	Traverses KML JSON structure to extract coordinate arrays from outerBoundaryIs/LinearRing/coordinates for polygons, and coordinates for points.
5	Convert to GeoJSON	Server-side GDAL (ogr2ogr) converts KML to standard GeoJSON FeatureCollection. Ensures CRS compliance with WGS84 (EPSG:4326).
6	Store in PostGIS	GeoJSON geometry inserted into PostgreSQL using ST_GeomFromGeoJSON(). Spatial indexes (GIST) applied for query performance.
7	Serve via REST API	GeoJSON FeatureCollection served via GET /api/schools/:id/geometry endpoint. Consumed by Leaflet and CesiumJS on the frontend.

4.1 KMZ Data Extraction — What the System Retrieves

- School boundary polygon (land demarcation / ubutaka)
- Building footprint polygons for each structure (classrooms, workshops, dormitories)
- Access road linestrings leading to the school
- Labeled point markers (entrances, water points, hazard zones)
- Overlaid drone orthomosaic imagery tiles (if embedded in KMZ)
- Metadata: placemark names, descriptions, timestamps, styles

4.2 PostGIS Spatial Schema

The core spatial tables in PostgreSQL + PostGIS:

- schools (id, name, district, sector, geom POINT, metadata JSONB)
- school_boundaries (school_id, geom POLYGON, source_kmz TEXT, uploaded_at TIMESTAMP)
- school_buildings (school_id, building_type TEXT, geom POLYGON, condition TEXT, area_m2 FLOAT)
- school_roads (school_id, geom LINESTRING, road_type TEXT, accessibility_score INT)
- population_snapshots (district TEXT, sector TEXT, population INT, fetched_at TIMESTAMP)
- decision_assessments (school_id, priority_level INT, recommendations JSONB, scored_at TIMESTAMP)

5. 3D Geospatial Visualization

Standard Google Maps provides 2D imagery that fails to convey terrain elevation, building height, slope risk, or spatial relationships between infrastructure and surrounding geography. This system implements a dual-mode visualization approach.

5.1 2D Dashboard Mode (Leaflet.js)

- Interactive tile-based maps with OpenStreetMap and satellite imagery basemaps
- GeoJSON polygon overlays for school boundaries, buildings, and risk zones
- Cluster markers for national school overview with drill-down capability
- Heatmap layers for population density and school coverage gaps
- Buffer zone circles (500m, 1km, 2km) to analyze school catchment areas

5.2 3D Terrain Mode (CesiumJS)

CesiumJS is the industry-standard open-source WebGL 3D globe engine, used by NASA, U.S. National Geospatial-Intelligence Agency, Bentley Systems, and thousands of government applications worldwide. It powers apps that reach millions of users and is released under Apache 2.0 license, making it free for government use.

- Full 3D globe rendering with real-world WGS84 terrain elevation (Cesium World Terrain)
- KML/KMZ files loaded natively via Cesium.KmlDataSource — school boundaries and buildings appear in 3D
- Drone orthomosaic imagery overlaid as XYZ tile imagery layer on the 3D globe
- Building extrusion: school structures rendered with configurable height based on floor count
- Terrain slope analysis: identify schools on steep gradients (>15° flagged as risk zones)

- Line-of-sight analysis: visibility between school and nearest road/health facility
- Camera flyTo animation: seamless navigation from national overview to individual school
- Shadow simulation based on sun position — assess natural lighting for classrooms

5.3 Drone Imagery Integration

Drone-captured orthomosaics (GeoTIFF format) are processed into tile pyramids (XYZ/TMS) using GDAL's gdal2tiles tool, then served as imagery layers in both Leaflet and CesiumJS. This delivers sub-10cm/pixel resolution compared to Google Maps' typical 15–50cm resolution for Rwanda, enabling:

- Precise roof condition assessment (cracking, corrosion, missing sheets)
- Accurate measurement of school grounds area (ubutaka)
- Detection of unauthorized structures or encroachments
- Tracking of construction progress over time through historical tile comparisons

6. Population Analytics & Decision Support Engine

6.1 Population Data Integration

The system integrates with the ArcGIS FeatureServer hosted by Esri (services5.arcgis.com), querying the population_count layer (FeatureServer/4) to retrieve GeoJSON population data for all administrative units across Rwanda. Data is matched to each school's sector and district coordinates using PostGIS ST_Within() spatial joins.

Population metrics computed per school:

- Total population within 500m, 1km, and 2km buffer zones
- School-age population (5–18 years) estimated using Rwanda national age distribution ratios
- Student-to-school ratio: how many school-age children exist per TVET seat available
- Population growth trend (year-over-year comparison when data refreshes)
- Underserved zone flag: sectors with population > 5,000 and no TVET school within 3km

6.2 Decision Support Engine — Scoring Model

The Decision Engine automatically evaluates each school against a weighted scoring matrix to produce a Priority Level and set of Recommendations.

Decision Factor	Weight (%)	Assessment Criteria
Infrastructure Condition Score	25%	Structural assessment (1–10 scale)
Age of School Buildings	15%	Years since last major renovation

Population-to-Capacity Ratio	20%	School-age population vs. current enrollment capacity
Terrain & Location Risk	20%	Slope grade, flood zone, landslide risk from terrain data
Compliance with RTB Standards	10%	Ratio of compliant facilities (toilets, labs, water, etc.)
Road Accessibility	10%	Distance from paved road, terrain accessibility index

6.3 Priority Classification Output

Priority Level	Score Range	Recommended Interventions
CRITICAL — P1	0–40	Immediate structural assessment; emergency renovation or relocation; suspend enrollment expansion
HIGH — P2	41–60	Schedule renovation within 6 months; apply for RTB/donor infrastructure funding; address safety deficiencies
MEDIUM — P3	61–75	Upgrade specific facilities; expand capacity if population ratio demands; improve access road
LOW — P4	76–100	Maintain current state; monitor annually; consider as model school for benchmarking

6.4 Decision Categories Supported

- School Renovation: triggers when structural score < 5 or buildings > 20 years old
- New Block/Room Construction: triggers when enrollment utilization > 90% and population demand is high

- School Relocation: triggers when terrain risk score is Critical and alternative land is available within 2km
- Upgrade to Higher Standard: triggers when school meets all basic standards but lacks advanced workshop or lab
- School Comparison: benchmarks a school against district or provincial averages on all metrics
- Catchment Gap Analysis: identifies sectors lacking TVET coverage relative to population density

7. Feasibility Study

7.1 Technical Feasibility

All core technologies proposed are production-proven, open-source, and actively maintained:

Technology	Maturity	Evidence of Readiness
CesiumJS	PROVEN	Used by NASA, NGA, Bentley; Apache 2.0; supports KML/KMZ natively since v1.7
PostGIS	PROVEN	Industry standard for spatial databases; used in GIS systems across 40+ countries
GDAL / ogr2ogr	PROVEN	Foundational GIS library; supports KML-to-GeoJSON conversion natively; widely deployed
React + TypeScript	PROVEN	Most widely used frontend framework; extensive GIS integration documentation
ArcGIS REST API	AVAILABLE	Rwanda population data available publicly at services5.arcgis.com; tested endpoint
JSZip + xml2js	PROVEN	Widely used for KMZ/KML parsing in browser and Node.js environments

VERDICT: Technically highly feasible. All components are mature, open-source, and have documented implementations in comparable government GIS systems across East Africa.

7.2 Operational Feasibility

- RTB's Digital Technologies Division (led by the Ag. Division Manager) provides an institutional home for the system
- LuxDev's ongoing 'Digital Skills for Quality TVET' project has already established data systems for monitoring in 11 schools across all 5 provinces — demonstrating appetite and capacity for digital tools
- Rwanda's existing TVET School GIS dashboard (ArcGIS Online) confirms precedent and infrastructure readiness
- RTB staff can be trained on the system's admin interface within 2-week onboarding sessions; no advanced GIS expertise required for day-to-day use

7.3 Financial Feasibility

The system leverages entirely open-source and freemium tools, minimizing licensing costs:

- CesiumJS: Free (Apache 2.0). Cesium ion optional for advanced terrain (freemium tier available)
- PostgreSQL + PostGIS: Free, open-source
- GDAL: Free, open-source
- Leaflet: Free, open-source
- ArcGIS Population API: Free public endpoint (no API key required for current endpoint)
- Drone image capture: One-time cost per school; can be phased by district priority

Primary costs are development labor, server infrastructure (cloud hosting), and drone imagery acquisition. These are manageable within RTB's existing SPIU budget envelope and eligible for development partner funding (LuxDev, Switzerland/PROMOST, KOICA frameworks).

7.4 Legal & Regulatory Feasibility

- Rwanda Civil Aviation Authority (RCAA) governs drone operations; RTB requires RCAA drone flight permits per site
- All data stored within Rwanda borders in compliance with Rwanda's Data Protection Law (Law No 058/2021)
- Population data sourced from public ArcGIS endpoint — no data sharing agreements required
- Open-source licenses (Apache 2.0, MIT) are compatible with government deployment

7.5 Risk Assessment

Risk	Level	Mitigation Strategy
Drone flight permit delays	MED	Engage RCAA early; prioritize highest-priority schools for initial drone surveys; use existing satellite imagery as interim fallback
Large KMZ file performance	LOW	Implement streaming KMZ processing; apply spatial indexing (PostGIS GIST); use 3D Tiles for large datasets in CesiumJS
Internet connectivity at schools	MED	System designed for central RTB use, not deployed at school level. Offline export capability added for field inspectors.
ArcGIS API availability	LOW	Cache population snapshots in local PostGIS DB; refresh on configurable schedule (weekly/monthly)
Data accuracy of school submissions	MED	Mandatory field validation in registration forms; RTB inspector sign-off workflow; audit trail for all submissions
Staff adoption resistance	LOW	Phased rollout with training; intuitive UI designed for non-GIS users; bilingual interface (English/Kinyarwanda)

8. Technical Implementation Plan

8.1 Development Phases

Phase	Title	Key Deliverables	Duration
Phase 1	Foundation & Data Architecture	Backend API scaffolding (NestJS), PostgreSQL + PostGIS setup, base school data schema, authentication (JWT + RBAC), environment configuration	3 Weeks
Phase 2	KMZ Processing Engine	KMZ upload endpoint, JSZip extraction, xml2js KML parser, GDAL GeoJSON conversion, PostGIS geometry storage, file version history	3 Weeks
Phase 3	School Registration & Data Forms	Complete school detail forms (React Hook Form + Zod validation), district/sector dropdowns from Rwanda admin boundaries, CRUD APIs, audit logging	2 Weeks
Phase 4	2D National & School Maps	Leaflet integration, national schools overview map, individual school 2D map with KMZ overlay, GeoJSON polygon rendering, buffer zones	3 Weeks
Phase 5	3D Visualization Module	CesiumJS integration, KmlDataSource rendering, drone orthomosaic tile overlay, building extrusion, terrain analysis, camera flyTo, slope classification	4 Weeks

Phase 6	Population Analytics Integration	ArcGIS FeatureServer API integration, population caching in PostGIS, ST_Within spatial joins, per-school population metrics calculation	2 Weeks
Phase 7	Decision Engine & Analytics	Weighted scoring algorithm, priority classification engine, recommendation generation, national analytics dashboard, charts (Recharts), PDF/Excel export	3 Weeks
Phase 8	QA, UAT & Deployment	End-to-end testing, RTB user acceptance testing (UAT), performance optimization, Docker containerization, cloud deployment (AWS/Azure), staff training	3 Weeks

Total Estimated Duration: 23 Weeks (~6 Months)	Delivery Model: Agile sprints (2-week cycles) with bi-weekly RTB review checkpoints

8.2 Recommended Technology Stack

Layer	Technology	Rationale
Frontend Framework	React 18 + TypeScript + Vite	Type safety, fast HMR, large ecosystem
UI Styling	TailwindCSS + Shadcn/UI	Rapid consistent design, accessible components
2D Mapping	Leaflet.js + React-Leaflet	Lightweight, fast, excellent Rwanda tile support

3D Mapping	CesiumJS (Apache 2.0)	Industry gold standard for 3D terrain + KML/KMZ
Backend Framework	Node.js + NestJS	TypeScript native, modular architecture, tested
GIS Processing	GDAL + ogr2ogr + JSZip + xml2js	Battle-tested KMZ/KML processing pipeline
Database	PostgreSQL 15 + PostGIS 3.4	Spatial queries, geometry storage, indexing
Charts	Recharts + D3.js	Composable, responsive, React-native charts
File Storage	AWS S3 / MinIO (self-hosted)	Drone tile storage, KMZ archives, reports
Containerization	Docker + Docker Compose	Consistent environments, easy deployment
Authentication	JWT + Role-Based Access Control	RTB admin, inspector, viewer roles

8.3 User Roles & Access Control

- RTB Administrator: Full access — create/edit/delete all schools, manage users, run decision engine, export reports
- District Inspector: Create/edit schools within their district, upload KMZ files, submit assessment forms
- RTB Analyst: Read-only access to all data, analytics dashboards, comparison tools, population data
- School Data Officer: Submit school registration forms and infrastructure updates for their assigned school only

9. Expected Outcomes & Impact

9.1 Quantitative Targets

Outcome Indicator	Baseline	Target (Year 2)
TVET schools with geospatial profiles in system	~0	100%
Schools with KMZ boundary data uploaded	~0	>85%
Infrastructure decisions supported by GIS data	<10%	>90%
Time to generate school condition report	2–3 weeks (manual)	<30 minutes (automated)
Schools meeting RTB quality standard (current 58%)	58%	>70% (via targeted interventions)
Sectors with identified TVET coverage gaps	Unknown	100% mapped and prioritized

9.2 Strategic Impact

- Transforms RTB from reactive to proactive infrastructure governance — from paper-based audits to real-time geospatial intelligence
- Enables evidence-based investment prioritization aligned with population demand — ensuring funds reach highest-need schools
- Provides Rwanda's Vision 2050 and NST-1 with a measurable TVET infrastructure monitoring mechanism
- Establishes Rwanda as a regional model for GIS-enabled education sector management in East Africa
- Accelerates the national target of achieving one TVET school per sector through data-driven gap analysis

9.3 Alignment with National Frameworks

- NST-1 Priority 4 — Quality Education: Direct infrastructure improvement mechanism
- ICT Sector Strategic Plan 2018–2024: Advances government digital transformation
- RTB Strategic Plan 2021–2024, Strategic Outcome 7: Provides the TVET Provision Mapping tool explicitly called for
- Rwanda Vision 2050 — Knowledge Economy: Builds digital capacity within the TVET sector

10. Conclusion

The RTB Schools GIS Monitoring & Intelligence System represents a transformative, technically sound, and operationally feasible investment in Rwanda's TVET infrastructure governance. By combining drone imagery, KMZ geospatial processing, 3D terrain visualization, and population analytics into a unified decision-support platform, RTB will gain unprecedented visibility into the condition, capacity, and risk profile of every TVET school across the country.

The system is built entirely on proven, open-source technologies — CesiumJS, PostGIS, GDAL, React — ensuring no vendor lock-in and minimal ongoing licensing costs. It aligns directly with RTB's strategic mandate, Rwanda's national development frameworks, and the growing body of GIS-enabled education management practice across Sub-Saharan Africa.

With a phased 6-month delivery plan, clear risk mitigations, and a rigorous decision engine grounded in infrastructure science and population data, this proposal offers RTB a concrete path from manual monitoring to intelligent, geospatial-powered governance of Rwanda's TVET school network.

Rwanda TVET Board | RTB Schools GIS Monitoring System | Project Proposal v1.0 | March 2026

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